

ATHUL S MENON

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SUMMARY

- Computer Science engineering student and full-stack developer focused on backend systems, AI-integrated applications, and modern web technologies. Experienced with React, Node.js, Java, Git, and Linux-based development environments. Currently learning Spring Boot.

EDUCATION

NSS College of Engineering

Bachelor of Technology in Computer Science Engineering (CGPA: 8.27)

Palakkad, Kerala

Aug 2023 – Present

National Higher Secondary School

Higher Secondary Education – Computer Science (Percentage: 96%)

Thrissur, Kerala

Mar 2020 – Mar 2022

National Higher Secondary School

Secondary School Leaving Certificate (Percentage: 100%)

Thrissur, Kerala

Mar 2019 – Mar 2020

EXPERIENCE

Derexa Labs Pvt. Ltd.

Software Engineering Intern

Remote

Jul 2026 – Present

- Contributed to an enterprise intelligence platform leveraging a graph-native, event-driven architecture to optimize cross-system queries and data processing.

PROJECTS

PhishGuard - AI Email Security & DLP System | React, Python, Flask, Scikit-learn, Docker Jan 2026 – Apr 2026

- Architected an AI-powered email security system using a 200-tree Random Forest classifier and TF-IDF vectorization to detect incoming phishing threats with text preprocessing and URL masking.
- Engineered a rule-based Multi-Stage Data Loss Prevention (DLP) engine executing regex pattern matching, confidential keyword detection, and domain verification to flag outgoing data leaks.
- Built a RESTful Flask API deployed on Hugging Face Spaces and integrated a React 19 frontend with real-time confidence scoring and dynamic endpoint routing.

PublicNote – Real-Time Collaborative Platform | React, Firebase, GitHub REST API, Vite, SCSS

Aug 2026

- Architected a real-time collaborative note and file-sharing platform using React and Firebase Realtime Database, enabling instant multi-user data synchronization across active rooms.
- Engineered a custom cloud file storage module integrating the GitHub REST API with Base64 encoding, chunked file uploads, and real-time progress tracking to handle diverse file formats.
- Optimized application performance and minimized database writes by implementing custom React hooks (useCallback, useRef), debouncing techniques, and Firebase OAuth session management.

Sports Attendance Tracker | Java 17, Spring Boot 3, React, PostgreSQL, Docker

2026

- Architected a full-stack platform using Java 17, Spring Boot 3, and React, implementing Spring Security 6 Role-Based Access Control (RBAC) and JWT/Session security for 3 user tiers.
- Engineered bidirectional state synchronization using TanStack Query to execute optimistic UI updates with automatic rollback on failure and client-side data caching.
- Designed a PostgreSQL database schema managed via Flyway SQL migrations and Spring Data JPA, ensuring database consistency and zero-downtime deployment pipelines.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML, CSS

Frontend: React.js, Tailwind CSS

Core Concepts: DSA & OOP

Backend: Node.js, Flask, Spring Boot, REST APIs & JWT

Databases & Tools: MongoDB, Firestore, Firebase, Git, GitHub, Linux, Docker

ACHIEVEMENTS & CERTIFICATIONS

Hacktoberfest 2025 Finisher | Open Source Community

Oct 2025

- Successfully completed by contributing 6+ meaningful pull requests to public open-source repositories.
- Collaborated with project maintainers while adhering to strict code review standards and Git/GitHub workflows.

Deep Learning Certification | IIT Ropar (NPTEL)

Jul 2025 – Oct 2025

- Completed an intensive 12-week course covering neural network architectures, optimization, and deep learning algorithms.

Google AI & Prompting Essentials | Google (Coursera)

Jul 2025

- Mastered prompt engineering techniques, contextual instruction design, foundational AI concepts, and practical integration strategies for responsible AI implementation.

LEADERSHIP & ACTIVITIES

- **Treasurer** – IEEE Student Branch, NSS College of Engineering
- **Chair** – IEEE Computer Society Chapter, NSS College of Engineering
- **Webmaster** – IEEE SIGHT Kerala Section